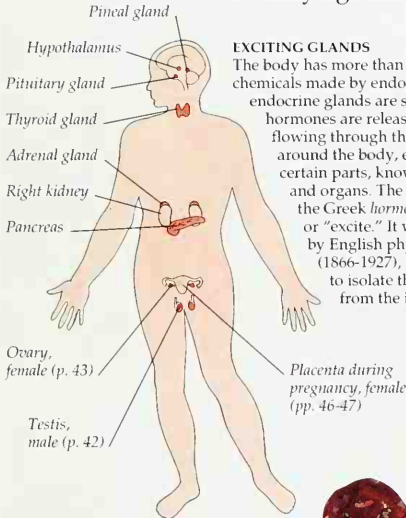


Body chemistry

REDUCE THE BODY TO ITS BASICS and it is like a complicated chemical factory. Every second, millions of substances take part in chemical reactions. The overall process of body chemistry is called metabolism. Physiologists are people who unravel its intricacies – how the organs and tissues function. Physiology is a complementary subject to anatomy, the study of structure. Muscles, bones, nerves, and guts all have characteristic chemical pathways. Three other chemical-based systems involve the whole body. These are the blood circulation (pp. 30-33), the endocrine or hormonal system, and the lymphatic system. Endocrinology is the study of hormones, the body's chemical messengers. It is a fairly recent science, begun in earnest only about a century ago. Lymph is a circulating fluid known as "blood's distant relative." It has only a vague system of vessels, and no pump of its own in the way that blood has the heart (pp. 28-29), but it is vital to defend the body against disease and infection.

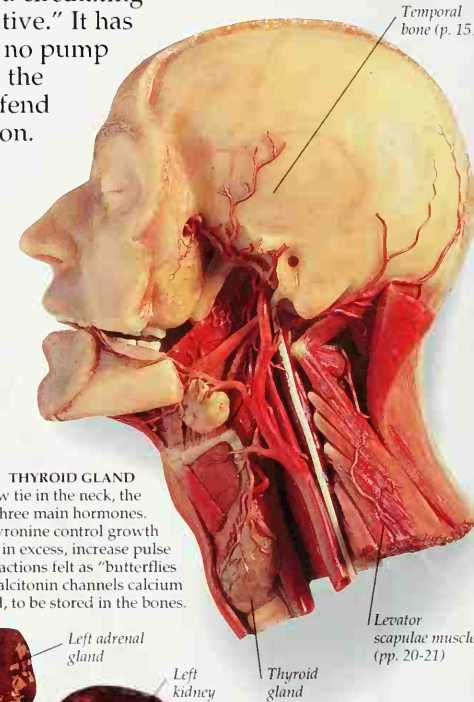
FIGHT OR FLEE

The hormone adrenaline prepares the body for physical action in the face of danger. It slows the digestive system (p. 34) and constricts blood vessels (pp. 50-51) to the skin to make more blood available to the muscles. The blood races and the muscles tense. This helps to fight the danger, in this case a bull, or flee from it.



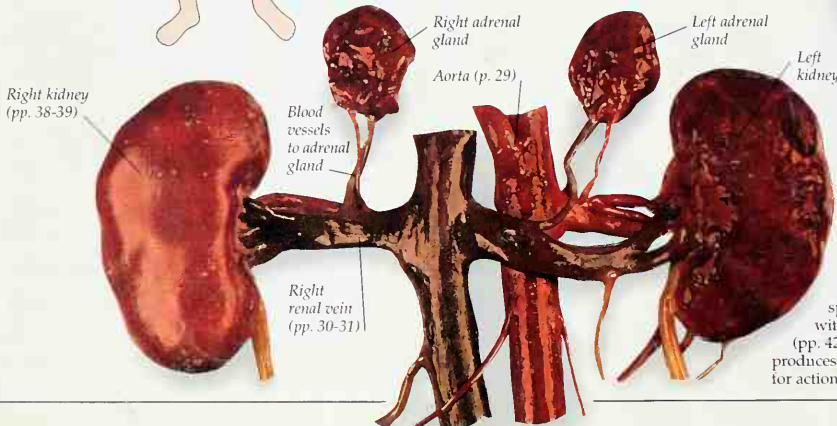
EXCITING GLANDS

The body has more than 50 hormones – messenger chemicals made by endocrine glands. The main endocrine glands are shown on the left. The hormones are released straight into the blood, flowing through the glands. They spread around the body, exciting or stimulating certain parts, known as their target tissues and organs. The word hormone is from the Greek *hormon*, meaning "to stir up" or "excite." It was suggested in 1905 by English physiologist Ernest Starling (1866-1927), three years after he helped to isolate the first hormone, secretin, from the intestine.



THYROID GLAND

Like an internal bow tie in the neck, the thyroid gland makes three main hormones. Thyroxine and triiodothyronine control growth and metabolic rate, and, in excess, increase pulse rate and intestinal contractions felt as "butterflies in the stomach." Calcitonin channels calcium from the blood, to be stored in the bones.



ADRENAL GLANDS

One adrenal, or suprarenal, gland sits on top of each kidney – another name for the kidney is the renal gland. The adrenal's outer part, the cortex, makes several hormones called corticosteroids. They include aldosterone to conserve salt and water in the kidneys, glucocorticoid steroids to speed up the metabolism and cope with stress, and the sex hormones (pp. 42-43). The inner part, or medulla, produces adrenaline, which readies the body for action in the fight-or-flight response.